

▼ Egg Gender Prediction Model - CORRECTED VERSION

Based on Lecturer Feedback: ARF Shafana Madam

Key Changes Made:

1. Random Forest max_depth: 10 → 3
2. Decision Tree max_depth: 10 → 3
3. Gradient Boosting max_depth: 10 → 3

Expected Result:

- Accuracy: ~99% (instead of suspicious 100%)
- No overfitting ☒

▼ SECTION 1: Install and Import Dependencies

```
[1]
✓ 4s # Install required packages
%pip install pandas numpy scikit-learn matplotlib seaborn openpyxl

... Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-packages (3.1.5)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.2.5)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl) (2.0.0)
```

```
[2]
✓ 2s # Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
import joblib
import warnings
warnings.filterwarnings('ignore')

print('All libraries imported successfully!')

... All libraries imported successfully!
```

SECTION 2: Mount Google Drive

```
[3]
✓ 33s # Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

SECTION 3: Configuration

```
[4]
✓ 0s # Dataset path in Google Drive
DATASET_PATH = '/content/drive/MyDrive/Egg Gender Prediction Model/Dataset/Egg_Dataset_Extended.xlsx'

# Model save path
MODEL_SAVE_PATH = '/content/drive/MyDrive/Egg Gender Prediction Model/'

# Model configuration
RANDOM_STATE = 42
TEST_SIZE = 0.2

print('Configuration complete!')
```

SECTION 4: Load Dataset

```
[5]
✓ 2s print('Loading dataset from Google Drive...')
df = pd.read_excel(DATASET_PATH)

print('Dataset loaded successfully!')
print(f'\nDataset shape: {df.shape[0]} rows x {df.shape[1]} columns')
print('\nFirst 5 rows:')
print(df.head())

print('\nColumn names:')
print(df.columns.tolist())

print('\nData types:')
print(df.dtypes)

print('\nMissing values:')
print(df.isnull().sum())

print('\nGender distribution:')
print(df['Gender'].value_counts())
```

*** Loading dataset from Google Drive...

Dataset loaded successfully!

*** Dataset shape: 1098 rows x 5 columns

First 5 rows:

	Egg Number	Width	Height	Shape Index(%)	Gender
0	2	40.52	54.06	74.95	Female
1	3	42.41	54.19	78.26	Unhatched
2	4	41.29	53.93	76.56	Female
3	5	37.16	54.40	68.30	Male
4	6	42.90	55.32	77.54	Female

Column names:

['Egg Number', 'Width', 'Height', 'Shape Index(%)', 'Gender']

Data types:

Egg Number	int64
Width	float64
Height	float64
Shape Index(%)	float64
Gender	object
dtype:	object

Missing values:

Egg Number	0
Width	0
Height	0

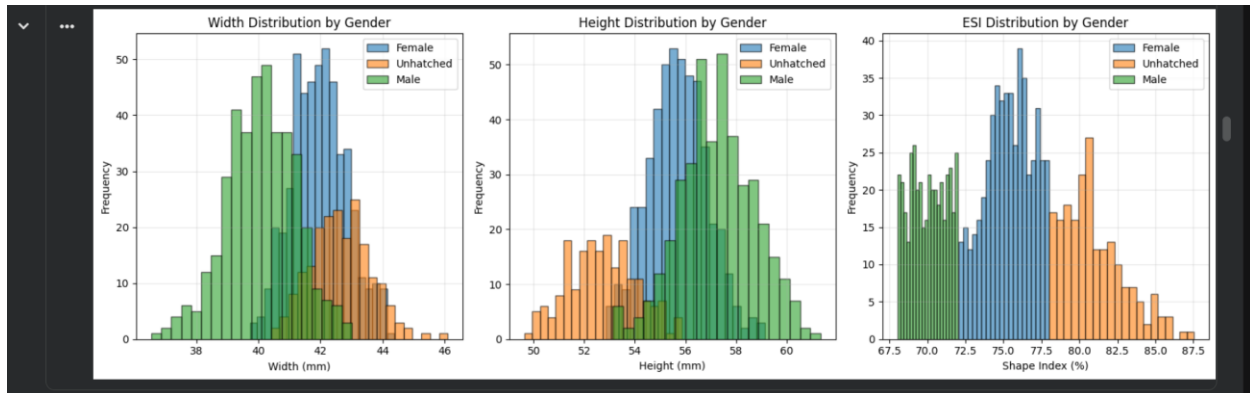
SECTION 5: Data Visualization

[6]
✓ Os

```
plt.figure(figsize=(15, 5))

# Width distribution
plt.subplot(1, 3, 1)
for gender in df['Gender'].unique():
    data = df[df['Gender'] == gender]['Width']
    plt.hist(data, alpha=0.6, label=gender, bins=20, edgecolor='black')
plt.xlabel('Width (mm)')
plt.ylabel('Frequency')
plt.title('Width Distribution by Gender')
plt.legend()
plt.grid(alpha=0.3)

# Height distribution
plt.subplot(1, 3, 2)
for gender in df['Gender'].unique():
    data = df[df['Gender'] == gender]['Height']
    plt.hist(data, alpha=0.6, label=gender, bins=20, edgecolor='black')
plt.xlabel('Height (mm)')
plt.ylabel('Frequency')
plt.title('Height Distribution by Gender')
```



[7]
✓ Os

```
X = df[['Width', 'Height', 'Shape Index(%)']].copy()

# Create additional features
X['Width_Squared'] = X['Width'] ** 2
X['Height_Squared'] = X['Height'] ** 2
X['ESI_Squared'] = X['Shape Index(%)'] ** 2
X['Width_Height_Interaction'] = X['Width'] * X['Height']

y = df['Gender']

print('Features created:')
print(X.columns.tolist())
print(f'\nFeature matrix shape: {X.shape}')

feature_columns = X.columns.tolist()
```

Features created:
['Width', 'Height', 'Shape Index(%)', 'Width_Squared', 'Height_Squared', 'ESI_Squared', 'Width_Height_Interaction']

Feature matrix shape: (1098, 7)

SECTION 7: Prepare Data

```
[8]
✓ Os
▶ label_encoder = LabelEncoder()
  y_encoded = label_encoder.fit_transform(y)

  print('Label Encoding:')
  for i, label in enumerate(label_encoder.classes_):
    print(f' {label}: {i}')

  X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded,
    test_size=TEST_SIZE,
    random_state=RANDOM_STATE,
    stratify=y_encoded
  )

  scaler = StandardScaler()
  X_train_scaled = scaler.fit_transform(X_train)
  X_test_scaled = scaler.transform(X_test)

  print(f'\nTraining set: {X_train.shape[0]} samples')
  print(f'\nTesting set: {X_test.shape[0]} samples')
```

Variables Terminal

✓ 11:51 AM Python

SECTION 8: Train Multiple Models - ✓ CORRECTED

CHANGES MADE:

- Random Forest max_depth: 10 → 3
- Decision Tree max_depth: 10 → 3
- Gradient Boosting max_depth: 10 → 3

```
[9]
✓ Os
▶ print('='*60)
  print('TRAINING MODELS WITH CORRECTED PARAMETERS')
  print('='*60)
  print('\nTraining models...\n')

  # ✓ CORRECTED: max_depth changed from 10 to 3
  models = {
    'Random Forest': RandomForestClassifier(
      n_estimators=100,
      random_state=RANDOM_STATE,
      max_depth=3 # ✓ CHANGED FROM 10 TO 3
    ),
    'Gradient Boosting': GradientBoostingClassifier(
```

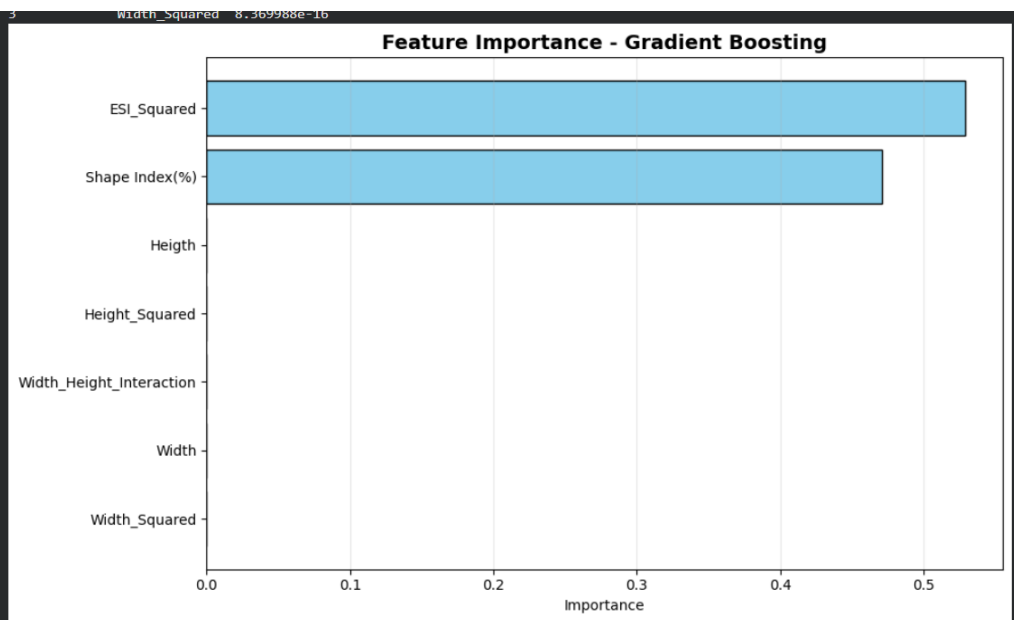
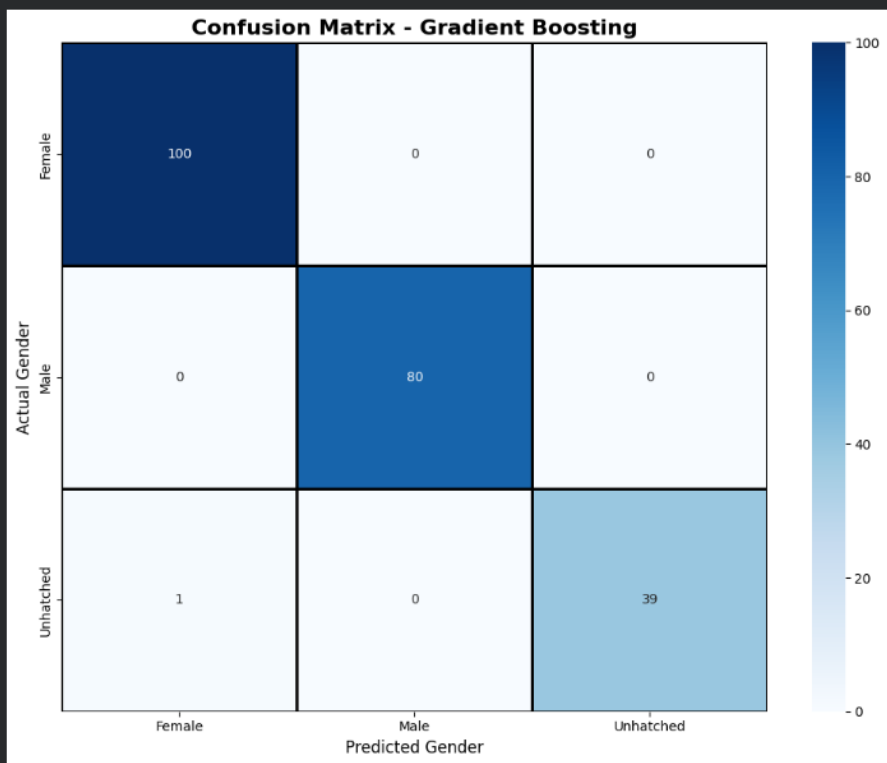
```
[9]
✓ Os
▶ print(f'\n✓ Best Model: {best_model_name} ')
  print(f'✓ Accuracy: {best_accuracy*100:.2f}%')
  print('='*60)

  ...
  TRAINING MODELS WITH CORRECTED PARAMETERS
  =====

  Training models...

  Training Random Forest... Accuracy: 99.09%
  Training Gradient Boosting... Accuracy: 99.55%
  Training Decision Tree... Accuracy: 99.55%
  Training SVM... Accuracy: 97.73%
  Training KNN... Accuracy: 98.18%
  Training Logistic Regression... Accuracy: 98.64%

  ✓ Best Model: Gradient Boosting
  ✓ Accuracy: 99.55%
  =====
```



... Sample Predictions:

Sample 1:

Width: 41.46 mm

Height: 55.35 mm

ESI: 74.90%

Actual: Female | Predicted: Female

✓ CORRECT

Sample 2:

Width: 41.16 mm

Height: 56.33 mm

ESI: 73.07%

Actual: Female | Predicted: Female

✓ CORRECT

Sample 3:

Width: 40.60 mm

Height: 54.99 mm

ESI: 73.82%

Actual: Female | Predicted: Female

✓ CORRECT

Sample 4:

Width: 40.95 mm

Height: 56.94 mm

ESI: 71.92%

Actual: Male | Predicted: Male

✓ CORRECT